PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

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PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

Practical no. 1

Porifera- sponges

Organism Name	Sponges
Kingdom	Animalia
Phylum	Porifera
Order	Dictyoceratida
Examples	Bath sponge (Spongia officinalis) light gray to black Calcareous Sponges. Calcareous sponges are the most primitive sponges, the only class of sponges that includes asconoid and synconoid varieties. Glass Sponges. Demospongiae.
Habitat	Aquatic, Freshwater, mostly marine
Nutrition	filter feeders (digest food within cells) Food passes from collar cells to food vacuoles to ameboid cells (Ameboid cells have two functions: engulf and digest food as well as secrete a material that helps to keep the sponge flexible)
Reproduction	both sexually and asexually. **asexually** by 1) external or internal buds 2) Fragmentation 2)regeneration 4)gemmule formation **Sexually**: The male sponge releases the sperm into the water which enters the body of the female sperm. Fertilization takes place inside the body of the female sponge after which larvae released to the water.
Movement	They are sessile
Body Parts	Vary in size, shape, color Multicellular but lack organized tissues Have cellular level of organization Outer layer of cells made of flattened epidermal cells (contractile fibers) Middle layer of cells is semifluid matrix (ameboid cells) Inner layer has flagellated cells called collar cells The flagella on collar cells produce water currents that flow through pores into central cavity out the osculum **Body types**: Sponges have three body types: asconoid, syconoid, and leuconoid. Asconoid sponges are tubular with a central shaft called the spongocoel. The beating of choanocyte flagella forces water into the spongocoel through pores in the body wall

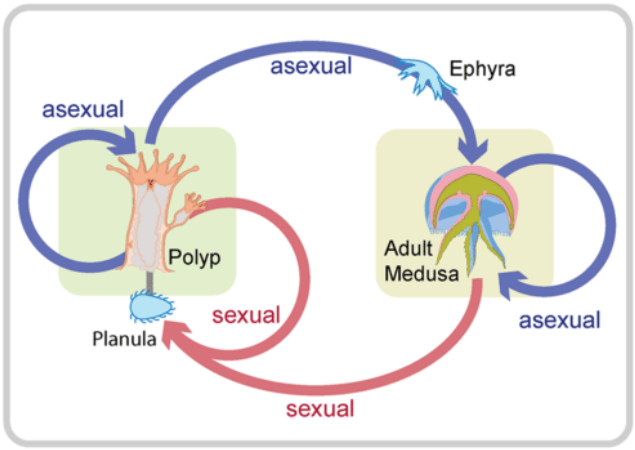
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PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

Practical no. 2

Cnidaria

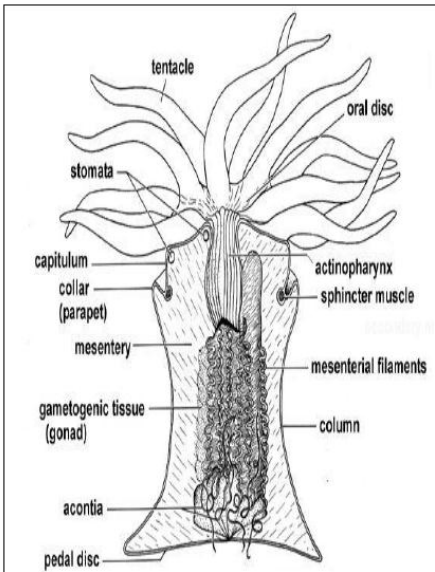
Cnidaria **due to the presence of cnidoblasts or cnidocytes on the tentacles and body surface**. They contain stinging capsules called nematocysts

Organism Name	Cnidarians
Kingdom	Animalia
Phylum	Cnidaria (also known as phylum Coelenterata)
Order	Semaeostomeae (jellyfishes) Anthoathecata (hydra)
Examples	**sea anemones** **jellyfishes**: Sea nettles (*Chrysaora fuscescens*) **hydra** (*Hydra vulgaris)* **corals**
Habitat	Except for hydra (freshwater organism), all the other cnidarians are marine
Nutrition	Cnidarians are carnivores, and some can also consume plant matter. **catch their food using their nematocysts or through filter feeding** Nematocysts threadlike fibers found in cnidocytes that are used to capture or sting prey
Reproduction	**asexual by budding or sexual using gametes** Depending on the species, cnidarians can be **monoecious** or **dioecious**. **both internal and external fertilisation** 
Movement	two body shapes: 1) **Polyp**: sessile as adult. 2) **Medusa**: motile **and swim by jet propulsion**
Body Parts	Radially symmetrical or spiral symmetry and have appendages that help them meet their nourishment two body shapes: 1) **Polyp**: They are sessile. They have arms and mouth at the top. 2) **Medusa**: They are motile with arms and mouth at the base. Multicellular, tubular, or bell-shaped animals found in shallow coastal waters tissue level of organization

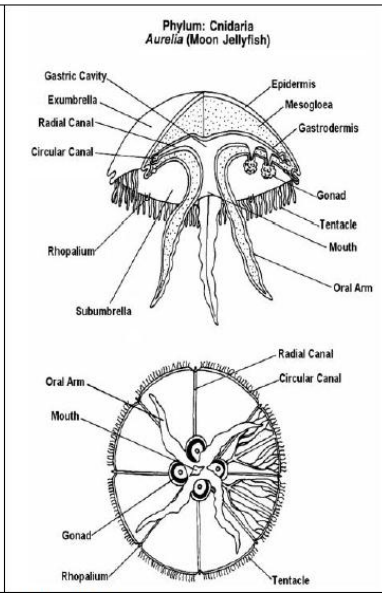
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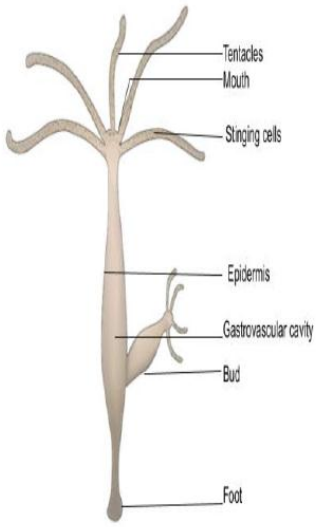
	Cnidocytes are specialized stinging cells Nematocysts threadlike fibers found in cnidocytes that are used to capture or sting prey



Sea anemone



Jelly fish



Hydra

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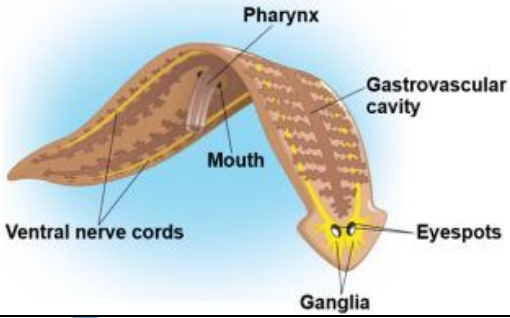
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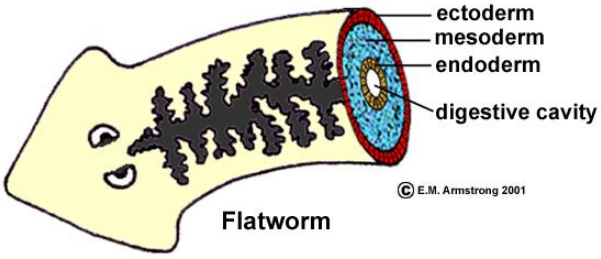
Practical no. 3

Platyhelminthes (Flatworms)

The phylum consists of four classes: **Trematoda (flukes), Cestoda (tapeworms), Turbellaria (planarians)**, and Monogenea

Organism Name	Flatworms
Kingdom	Animalia
Phylum	Platyhelminthes
Order	Plagiorchiida Cyclophyllidea
Examples	**Fluke:** common liver fluke (*Fasciola hepatica*) **Tape worm**: beef tapeworm (*Taenia saginata*)
Habitat	Live in moist areas or in aquatic environments or inside a host
Nutrition	**free-living heterotrophs or parasites** **absorb nutrients by consuming them through a mouth**.
Reproduction	**sexually** Flatworms are hermaphroditic (having both male and female sex organs) **Asexually** Through a process called **"fission,"** **planarians** can reproduce **asexually by simply tearing** themselves into two pieces -- a head and a tail
Movement	flatworm moves is through the **use of cilia** that are located on the ventral surface epidermis **larger** species use **muscular movements** of their entire body to creep, swim, twist
Body Parts	Flattened bodies; Bilateral symmetry • **Head region** with eye spots (see light/dark not images) in many • **Simple brain**; soft bodies • **Incomplete digestive** tract, meaning they eat and dispose of solid wastes through the same opening.



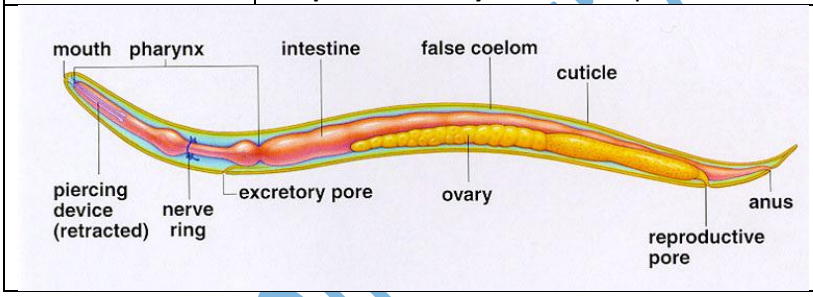


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Practical no. 4

Nematoda (Round worms)

Organism Name	**Roundworm and Hookworm**
Kingdom	Animalia
Phylum	Nematoda
Order	Ascaridida (Roundworm) Rhabditida (Hookworms)
Examples	**Roundworm: Ascaris lumbricoides** **Hookworm: Necator americanus** **(**commonly known as the New World hookworm**)**
Habitat	**round worms :freshwaters, soils, and marine** **Hookworms** mostly in **areas with moist, warm climates**
Nutrition	**Parasitic (feed in the body of host)**
Reproduction	Roundworms **reproduce sexually**. Sperm and eggs are produced by separate male and female adults. Fertilization takes place inside the female organism. ... The eggs hatch into larvae, which develop into adults. **Round worm:** Uteri may contain up to 27 million eggs at a time, with 200,000 being laid per day **Hook worm:** They can produce between 5,000 and 10,000 eggs per day
Movement	move by **contraction of the longitudinal muscles**
Body Parts	**body cavity** with internal organs **complete digestive system** with mouth and anus **simple nervous system** with a primitive brain
	

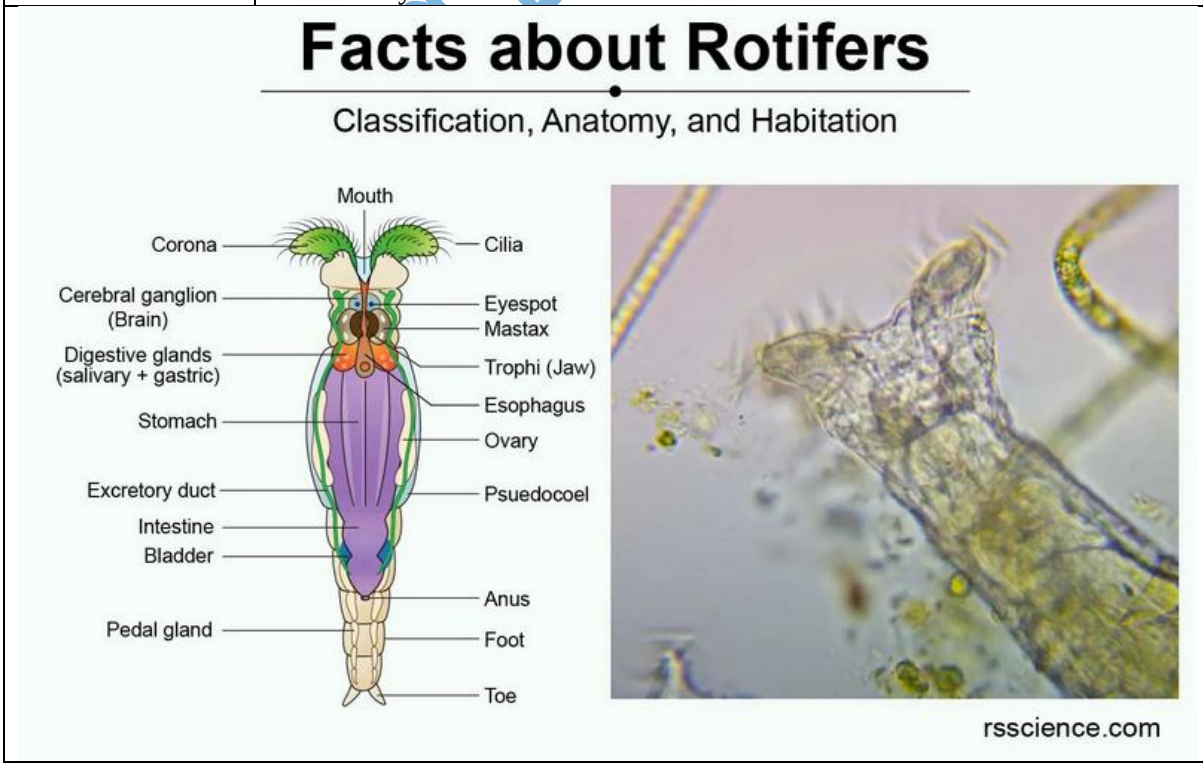
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Practical no. 5

Rotifers : Brachionus plicatilis

Organism Name	Rotifers (**wheel animalcules)**
Kingdom	Animalia
Phylum	Rotifera
Order	Ploima
Examples	**Rotifers** Brachionus quadridentatus. Colurella adriatica. Habrotrocha angusticollis. **Cephalodella gibba**. Brachionus urceolaris. Lepadellaacuminata.
Habitat	**brackish waters** freshwater broad distribution in salt lakes
Nutrition	feed on **smaller phytoplankton, decomposing organic materials, and unicellular algae** raised in the aquaculture industry as food for fish larvae
Reproduction	Having separate genders, their reproduction is sexual. A few animal categories contain just females. They reproduce by parthenogenesis. High rate of reproduction (every 4 hours female eggs)
Movement	Move by ciliated organ called corona helps in swimming some sessile
Body Parts	crown of cilia at the head-enveloping mouth external layer of chitin



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PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

Practical no. 6

Annelids (Segmented Worms)

Organism Name	Annelids
Kingdom	
Phylum	Annelida
Order	Opisthopora (Earthworm) Arhynchobdellida (Leeches)
Examples	**Earthworm: *Amynthas mekongianus*** **Leeches: *Hirudo medicinalis***
Habitat	**Earthworm** generally inhabit pastureland burrows into the muddy banks of the River live **anywhere there is moist soil and dead plant material** **Leaches** freshwater, terrestrial and marine.
Nutrition	**Earthworms** feed on organic matter found in the soil such as dead leaves and small soil particles that are digested in the intestine. Some species of **leech** (hematophagus) feed on blood. Other species feed on decomposing bodies and open wounds of amphibians, reptiles, fish and even mammals.
Reproduction	**(Sexually)** Earthworms are **hermaphrodite** organisms, meaning that each earthworm has **both male and female sexual reproduction** organs. Fertilization is external and occurs **in cocoon**. **Asexually** by regeneration. **(Only Sexually) Medicinal leeches** are hermaphrodites that reproduce by sexual mating, laying eggs in clutches of up to 50 near (but not under) water, and in shaded, humid places.
Movement	**Earthworm** moves **using circular and longitudinal muscles**, as well as bristles called setae. The earthworm can push the setae out of its body to grab the soil around it. To move forward, the worm uses its setae to anchor the front of its body and contracts the longitudinal muscles to shorten its body. Leeches move with the help of the anterior and posterior suckers and longitudinal muscles along the length of the body
Body Parts	The two ends of an earthworm: -Anterior end (head end) and Posterior end (tail/rear end) -Clitellum [a form on the head end and holds the reproductive part (cocoons form)] Body parts on an earthworm: -Mouth (right under the prostomium) -Prostomium (flap of skin/"upper lip") -Segment (muscles) -Setae (for movement) -Anus (the last segment, has an opening for waste)

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PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

Practical no. 7

Mollusks (phylum Mollusca)

Organism Name	Mollusks
Kingdom	
Phylum	**Phylum Mollusca**
Order	Ostreida (Oyster) Stylommatophora (Snails) Octopoda (Octopus)
Examples	The phylum Mollusca consists of bivalves, octopus, snails and slugs, squid, and the chambered nautilus. **Oyster: *Ostrea angasi*** **Snail: *Helix pomatia*** ***(***Roman snail) **Octopus: *Octopus vulgaris***
Habitat	**Oyster** This species is found in sheltered, silty or sand-bottomed estuaries at depths between 1 and 30 metres. **Snail** lives in forests, open habitats, gardens, and vineyards, especially along rivers prefers high humidity and lower temperatures **Octopus** worldwide distribution in tropical, subtropical and temperate waters throughout the world. They prefer the floor of relatively shallow, rocky, coastal waters, often no deeper than 200 m
Nutrition	**Oyster** filter feeders, feeding on, and taking in anything small enough to be filtered in their gills. This may include plankton, microalgae or inorganic material. **Snail** Snails and slugs eat **with a jaw and a flexible band of thousands of microscopic teeth**, called a radula. The radula scrapes up, or rasps, food particles and the jaw cuts off larger pieces of food, like a leaf, to be rasped by the radula. **Octopus** The common octopus hunts at dusk. Crabs, crayfish, and bivalve molluscs (such as cockles) are preferred, eats almost anything it can catch. able to change colour to blend in with its surroundings able to jump upon any unwary prey Using its beak, it is able to break into the shells of shelled molluscs.
Reproduction	**Oyster** Oysters are not exactly male or female. They are protandric which means they can change from male to female over their life. During an oysters spawning season, the young male releases his sperm and the older females release their eggs. **Snail** Some snails are hermaphrodites, some reproduce sexually and some freshwater pond species reproduce asexually (parthenogenesis). **Octopus** octopus **reproduces sexually** Typically, males die within months after mating, while females watch over their eggs until they hatch and then die shortly after.
Movement	**Oyster**

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	Snail snails and slugs propel themselves by generating a **series of muscular pulses on their feet** The foot provides wave-like movements which propel the body of the snail. **Octopus** Octopus move using **a form of jet propulsion**. Water is forced out of a special structure called a funnel, and the octopus is propelled in the opposite direction
Body Parts	sturdy limestone shells bilateral symmetry inhale through gills or lungs (situated underneath a defensive layer called the mantle) possess a rasping, tongue-like radula for scratching sustenance (except bivalves) circulatory framework includes a three-chambered heart and open-streaming framework (except octopuses and squids)

Practical no. 8

Arthropods (Chelicerata)

subphylum Chelicerata within arthropoda

Organism Name	Insects
Kingdom	
Phylum	Arthropoda
Order	Xiphosura (Horseshoe) Araneae (Spider) Trombidiformes (mites) Ixodida (Ticks)
Examples	spiders, scorpions, harvestmen, mites and ticks **Horseshoe Crab: *Limulus polyphemus*** **Spiders: *Plexippus petersi*** **Mites: *Trombidium holosericeum*** **Ticks: *Ixodes ricinus***
Habitat	the Chelicerata are found primarily in **terrestrial environments** worldwide (except **marine horseshoe crabs**) The only class within Chelicerata which is **not entirely marine** is the Arachnida.
Nutrition	**Horseshoe crabs**: Horseshoe crabs are omnivorous scavengers, feeding upon **small bivalves, mollusks, worms, dead fish and algae**. **Spider:** This spider is commonly known as the tropical flycatcher or small zebra jumper. **Mites:** they are seen as red globules on their hosts, sucking body liquid without severely harming the host. **Ticks:** Ticks are **obligate blood feeders**. the tick-host parasitic interaction is complex

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PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

Reproduction	**Horseshoe crabs**: Sexual reproduction. On the beach, the females dig small nests and deposit eggs, then the males fertilize the eggs **Spider:** This spider is commonly known as the tropical flycatcher or small zebra jumper. **Mites:** they are seen as red globules on their hosts, sucking body liquid without severely harming the host. **Ticks: reproduce sexually,** A female lays several thousand eggs at a time **asexual process** called parthenogenesis, meaning female ticks can reproduce without a male.
Movement	**Horseshoe crabs**: The horseshoe crab has a hard exoskeleton and 10 legs, which it uses for walking along the seafloor. posterior four pairs of appendages are normally used for walking **Spider:** The legs of spiders move using **a combination of the hydraulic pressure of body fluid and muscle flex** **Mites:** **peristaltic mode** of motion **Ticks:** **Ticks can only crawl**; they cannot fly or jump
Body Parts	segmented body segmented limbs exoskeleton (a thick chitinous cuticle) Chelicerates have two body segments; a cephalothorax and an abdomen no antennae, but have six pairs of appendages anterior appendages are called the chelicerae posterior four pairs of appendages

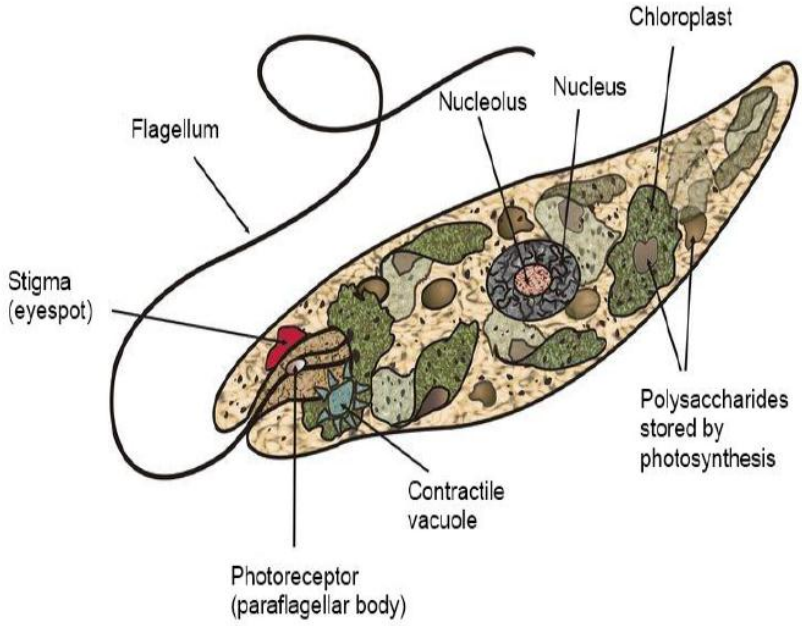
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Practical No.

Euglena

Organism Name	
Kingdom	Protista
Phylum	Euglenoza
Order	Euglenales
Examples	
Habitat	freshwater, saltwater, marshes and also in moist soil
Nutrition	characteristics of both plants and animals. manufacture their own food also consume food
Reproduction	**asexually by binary fission**
Movement	capable of movement by flagellum at night
Body Parts	

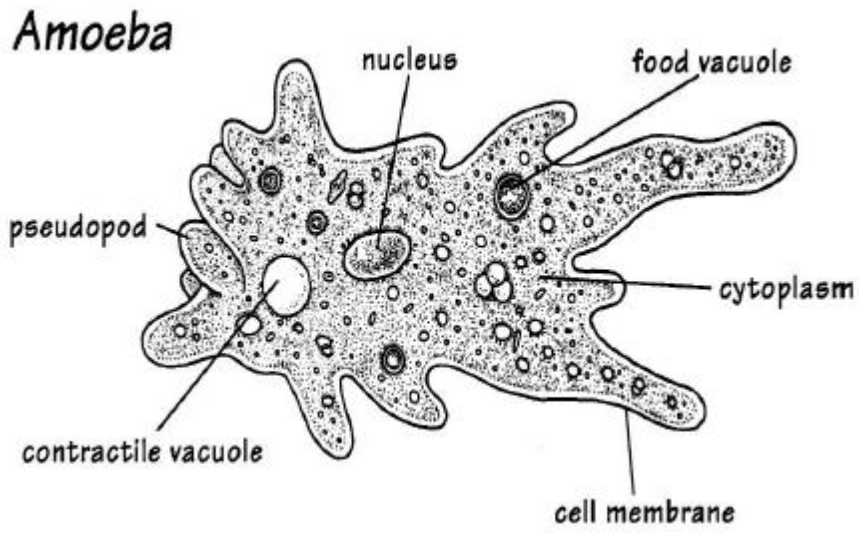


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Practical No.

Amoeba

Organism Name	
Kingdom	Protista (Protozoa)
Phylum	Amoebozoa
Order	Euamoebida
Examples	
Habitat	freshwater pools, ponds, lakes, streams
Nutrition	consumes food either through the process of phagocytosis or pinocytosis
Reproduction	reproduction is through asexual means, such as binary fission
Movement	move in a primitive crawling manner (by using temporary "false feet" known as pseudopods)
Body Parts	single celled organisms false feet called pseudopodia Structure---3 parts the cytoplasm -- 2 layers – the outer ectoplasm and the inner endoplasm plasma membrane -- very thin, double-layered membrane composed of protein and lipid molecules the nucleus other cellular organelles such as a contractile vacuole, mitochondria, Golgi apparatus and fat globules



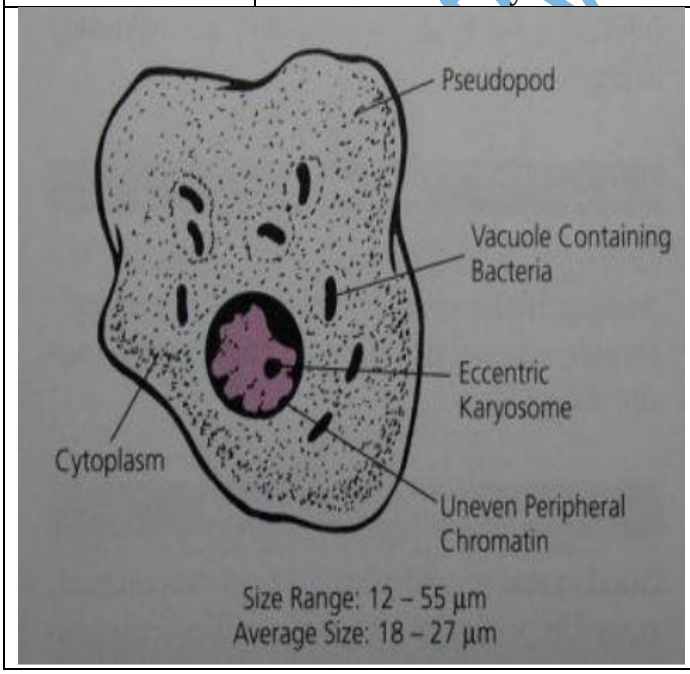
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PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

Practical No.

Study of Entamoeba

Organism Name	
Kingdom	
Phylum	Amoebozoa
Order	Amoebida
Examples	
Habitat	present in large intestine
Nutrition	Trophozoites are anaerobic parasite, It is the growing and feeding stage of parasite The trophozoite (feeding-dividing form) is approximately 10-20µm in diameter and feeds primarily on bacteria. Ingested RBCs and food materials present in endoplasm
Reproduction	**through binary fission**
Movement	Trophozoites are actively motile with the help of pseudopodia
Body Parts	Small cells **Sape;** not fixed because of constantly changing position **Size:** ranging from 18-40 µm; average being 20-30 µm **Cytoplasm:** two portion; a clear transparent ectoplasm and a granular endoplasm Ingested RBCs, tissue granules and food materials are also found in endoplasm **Nucleus:** It is single, spherical shape and size ranging from 4-6µ Nucleus contains central karyosome and fine peripheral chromatin.



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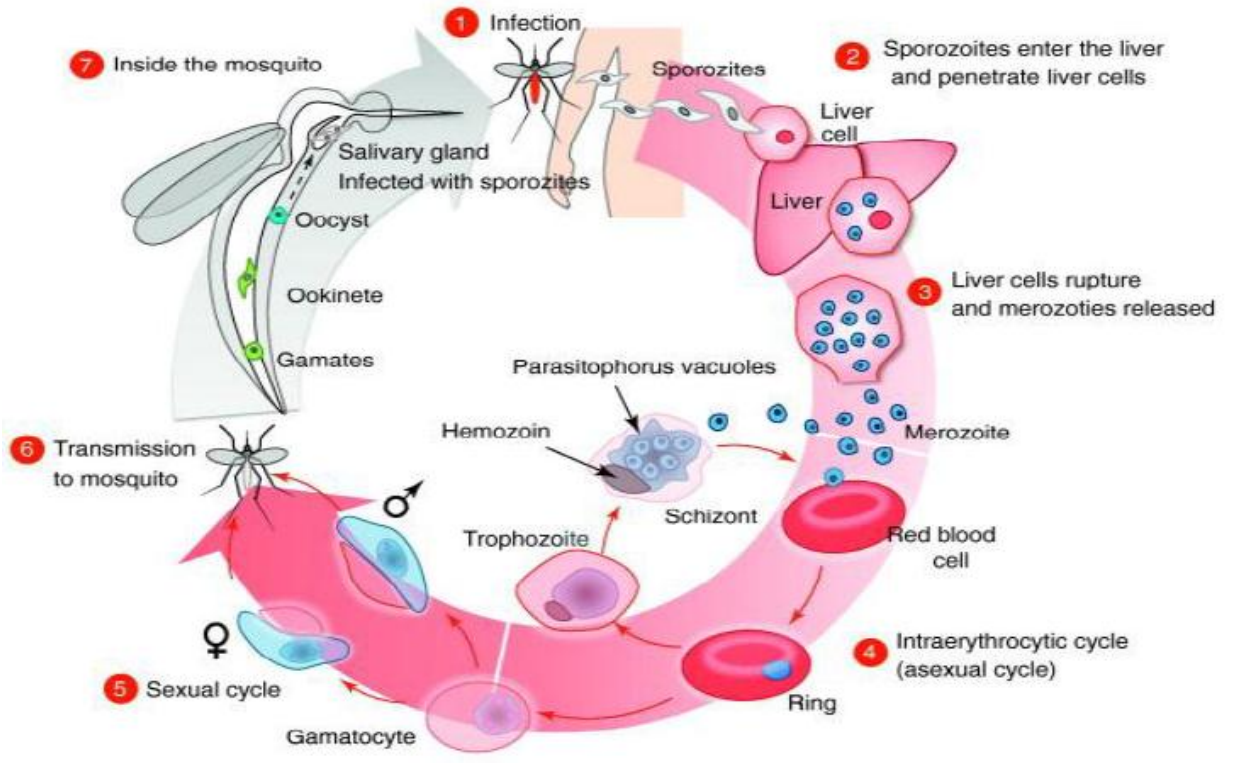
PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

Practical No.

Study of Plasmodium

genus of the class of Sporazoa that includes the parasite that causes malaria.

Organism Name	
Kingdom	Class sporozoa Type of protozoa
Phylum	Apicomplexa
Order	Haemosporida
Examples	
Habitat	Live in the body of host
Nutrition	They are obligate parasites of insects (such as mosquitoes) and vertebrates
Reproduction	reproduces by multiple fissions able to divide only within a host cell.
Movement	Glow gliding (with special organs) they secrete sticky membrane-spanning adhesins
Body Parts	single-celled organism plasma membrane nucleus mitochondria ribosome dense granules apical end



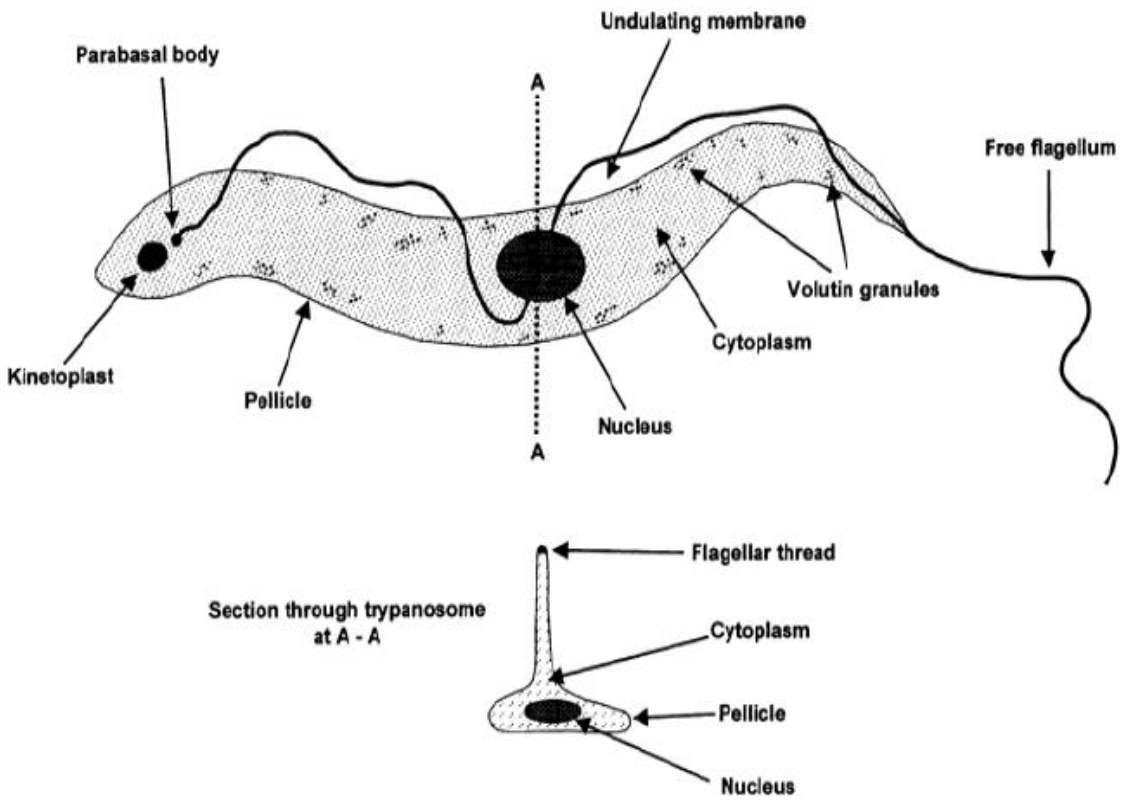
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Practical No.

Study of Trypanosoma

Organism Name	
Kingdom	
Phylum	Euglenozoa
Order	Trypanosomatida
Examples	
Habitat	
Nutrition	Heterotroph (parasite) absorbing nutrients, through their outer membrane, from the body fluids of the host. transmitted by the tsetse fly
Reproduction	**Multiply by binary fission in the body fluid of host**
Movement	by movement of the undulating membrane and the free flagellum (when present)
Body Parts	common characteristics with other members of the phylum Euglenozoa, particularly the stiffening paraxial rod in the flagellum, and characteristics common to the order Kinetoplastida, specifically a large clump of DNA located at one end.

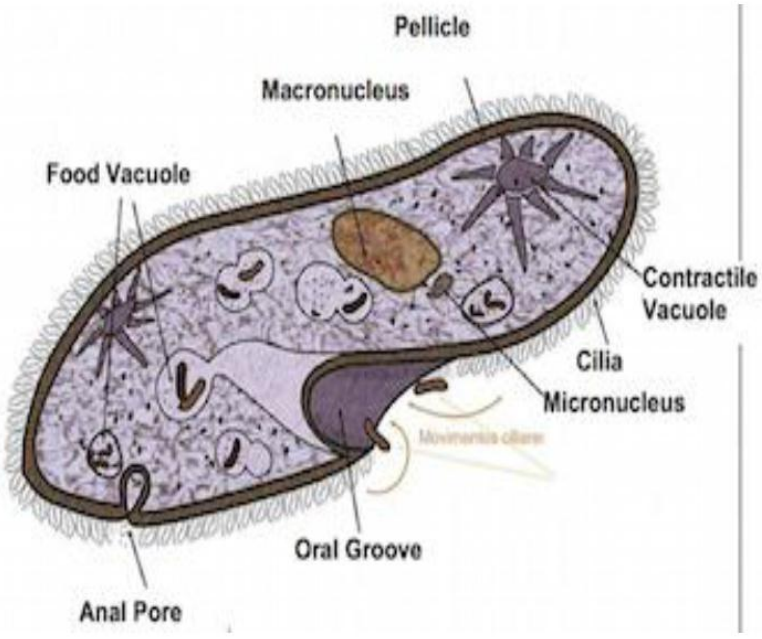


PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

Practical No.

Study of Paramecium Caudatum.

Organism Name	
Kingdom	
Phylum	Ciliophora
Order	Peniculida
Examples	
Habitat	freshwater ponds, pools, ditches, streams, lakes, reservoirs and rivers Paramecia are holozoic, living on bacteria, algae, and other small organisms
Nutrition	Paramecia are holozoic, living on bacteria, algae, and other small organisms. Cilia in the oral groove sweep food particles in the water into the cytostome, from which they are carried into the cytopharynx by the undulating membrane
Reproduction	**asexual, by binary fission (characterized as "the sole mode of reproduction in ciliates"**
Movement	locomotion mainly occurs by movement of cilia. It can move forward and backward.
Body Parts	slipper shaped. 150 to 300 µm in length blunt anteriorly somewhat pointed posteriorly. asymmetrical appearance The pellicle is a clear, elastic membrane that may be ornamented by ridges or papilla-like projections cilia arranged in lengthwise rows. Just below the pellicle is the thin clear ectoplasm that surrounds the larger mass of granular endoplasm food vacuoles containing food two contractile vacuoles two nuclei: a large kidney-shaped macronucleus and a smaller micronucleus



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PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

Practical No.

Phylum Echinodermata

Organism Name	Echinoderms
Kingdom	
Phylum	**phylum Echinodermata**
Order	Paxillosida
Examples	Starfish: Astropecten aranciacus Sea cucumbers: Actinopyga echinites sea urchins
Habitat	exclusively marine animals
Nutrition	Nutrition: Echinoderm feeding depends on the class and species filter feeders deposit feeders
Reproduction	**They replicate agamically by a discontinuity or also sexually with outside fertilization** **Sexes are separate.** **Fertilization is external.** **Lost parts can be regenerated**
Movement	use spines and tube feet , may move forward with any area of the body and reverse direction without turning around.
Body Parts	The larval forms show bilateral symmetry and adult forms show radial symmetry They are triploblastic. (body derived from three embryonic cell layers (ectoderm, mesoderm, and endoderm) □ It exhibits organ system grade of organisation. (simple digestive system with a mouth, stomachs, intestineand anus) □ They have a true coelom. (make up the five-area symmetry.) □ The body is uniquely shaped. It can star like, elongated or spherical. □ The body is unsegmented without a head. □ The body surface is covered with calcareous spicules. □ Body cavity has the distinguishing water vascular system. □ Tube feet help in locomotion. □ The brain is absent but a nervous system is present with a nerve ring and radial nerve cords. □ Respiration occurs through tube feet and gills. □ Sense organs are poorly developed and include tactile organs, chemoreceptors, terminal tentacles etc.

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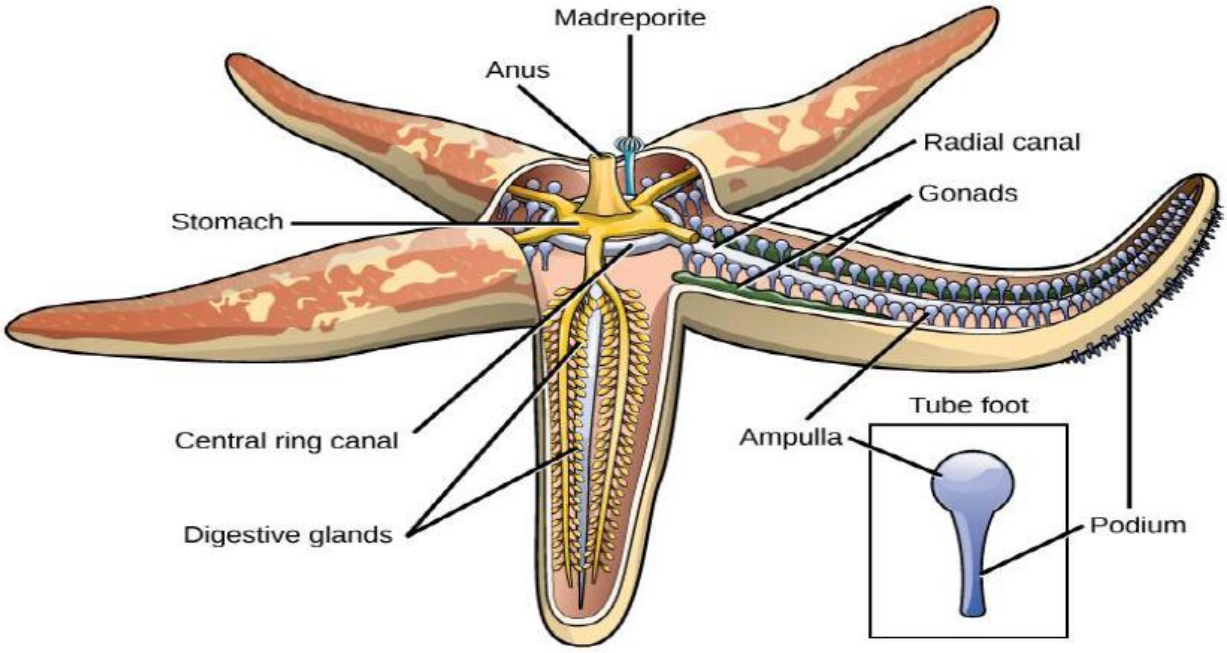


Figure: Structure of star fish

Plasmodium falciparum

Background	Falciparum malaria was familiar to the ancient Greeks It was the Romans who named the disease "malaria"—*mala* for bad, and *aria* for air Johann Friedrich Meckel, must have been the first to see *P. falciparum* It is derived from the Latin *falx,* meaning "sickle" and *parum* meaning "like or equal to another"
Route of Cause (e.g. infectious etc.)	transmitted through the bite of a female Anopheles mosquito *P. falciparum* is now generally accepted to have evolved from *Laverania* (a subgenus of *Plasmodium* found in apes) species present in gorilla in Western Africa
Parasitic Nature (e.g zoonotic etc)	Zoonotic (*P. falciparum* is now generally accepted to have evolved from *Laverania* (a subgenus of *Plasmodium* found in apes) species present in gorilla in Western Africa) Infectious disease
No. of Hosts	two (primary insect and secondary mammals (vertebrates))
Morbidity Rate Mortality Rate	According to the latest World malaria report, there were 241 million cases of malaria in 2020 compared to 227 million cases in 2019. The estimated number of malaria deaths stood at **627 000 in 2020** – an increase of 69 000 deaths over the previous year. It is responsible for around 50% of all malaria cases deadliest parasite in humans Malaria is the 1st leading cause of mortality due to parasitic disease in humans.

Compiled By: Atiq

PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

Entamoeba histolitica

Background	discovered in Russia in 1873 by Friedrich Losch His early observations came from the case of a young farmer who had been suffering from chronic dysentery histolytica never developed symptoms and spontaneously clear the infection, ie those who were shedding E. hystolitica
Route of Cause (e.g. infectious etc.)	the causative agent of human amoebiasis The parasite lives only in humans and is passed in the feces (poop) of an infected person. A person gets amebiasis by putting anything in their mouth that has touched infected feces or by eating or drinking food or water contaminated with the parasite.
Parasitic Nature (e.g zoonotic etc)	Parasitic Infections an **anaerobic parasitic amoebozoan** invading parasite: histolytica from fecally contaminated food or water **leads to intestinal amoebiasis**. The amoebae then invade through the intestinal mucosa, causing amoebic colitis and bloody diarrhea. They can invade through the portal circulation and cause liver abscesses.
No. of Hosts	Humans are the host of infection
Motility Rate Morbidity Rate	Worldwide, approximately 50 million cases of invasive E histolytica disease occur each year, resulting in **as many as 100,000 deaths**. This represents the tip of the iceberg because only 10%-20% of infected individuals become symptomatic. The incidence of amebiasis is higher in developing countries. histolytica is **close to 1% worldwide**. Entamoeba histolytica is the second leading cause of mortality due to parasitic disease in humans.

Visceral leishmaniasis

Background	History. Kala-azar first came to the attention of Western doctors in 1824 in **Jessore, India (now Bangladesh)**, where it was initially thought to be a form of malaria. Assam gave kala-azar one of its common names, Assam fever
Route of Cause (e.g. Aerosol, infectious etc.)	*Leishmania* parasites are transmitted by the bite of infected phlebotomine sand flies
Parasitic Nature (e.g zomotic etc)	human leishmaniasis can have zoonotic or anthroponotic transmission patterns
No. of Hosts	Two (phlebotomine sand flies and human)
Motility Rate Morbidity Rate	It is estimated that between 0.9 and 1.7 million people are newly infected every year, but only a small fraction of them will develop the disease and 20,000–30,000 will eventually die

Compiled by: Tariq Rehmani

PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

Trypanosoma

Background	In 1841, **Gabriel Valentin found** flagellates that today are included in Trypanoplasma in the blood of trout. observation of trypanosomes in human blood was made by **the British Colonial surgeon Robert Michael Forde** (1861–1948) in 1901 when he examined a steamboat captain in The Gambia
Route of Cause (e.g. Aerosol, infectious etc.)	They are transmitted to humans by tsetse fly (Glossina genus) bites which have acquired their infection from human beings transmitted by arthropod vectors and to mammalian hosts
Parasitic Nature (e.g zomotic etc)	Zoonotic transmission
No. of Hosts	Two (arthropod and mammalian)
Motility Rate Morbidity Rate	When left untreated, the mortality rate of African sleeping sickness is **close to 100%**. It is estimated that 50,000 to 500,000 people die from this disease every year.

Compiled By: Atiq Rehman

Microcopy:

Microscopy is the act of using a microscope to view tiny things that cannot be seen with the unaided eye. There are three main types: **optical microscopy, scanning probe microscopy, and electron microscopy**

A **microscope** is an instrument that magnifies objects otherwise too small to be seen, producing an image in which the object appears larger.

Most photographs of cells are taken using a microscope can also be called **micrographs**.

just one lens, they are called **simple microscopes**.

compound microscopes, meaning that they have multiple lenses.

Magnification is a measure of how much larger a microscope (1500X) (0.2 micro meter)

The **resolution** of a microscope or lens is the smallest distance by which two points can be separated

Both magnification and resolution are important if you want a clear picture of something very tiny.

Electron microscopes differ from light microscopes in that they produce an image of a specimen by using a **beam of electrons** rather than a beam of light. Electrons have much a shorter wavelength than visible light, and this allows electron microscopes to produce higher-resolution images than standard light microscopes. Electron microscopes can be used to examine not just whole cells, but also the subcellular structures and compartments within them.

There are two major types of electron microscopy.

PRACTICAL POINTS FOR ZOO-201 AND ZOO-103

scanning electron microscopy (**SEM**)

In **transmission electron microscopy** (**TEM**)

- TEM: magnifies 50 to ~50 million times; the specimen appears flat
- SEM: magnifies 5 to ~ 500,000 times; sharp images of surface features
- STEM: magnifies 5 to ~50 million times; the specimen appears flat

resolution of 0.2 nm

How to Use a Microscope

Compound Microscopes

Turn the revolving turret (2) so that the lowest power objective lens (eg. 4x) is clicked into position.

1. Place the microscope slide on the stage (6) and fasten it with the stage clips.
2. Look at the objective lens (3) and the stage from the side and turn the focus knob (4) so the stage moves upward. Move it up as far as it will go without letting the objective touch the coverslip.
3. Look through the eyepiece (1) and move the focus knob until the image comes into focus.
4. Adjust the condenser (7) and light intensity for the greatest amount of light.
5. Move the microscope slide around until the sample is in the centre of the field of view (what you see).
6. Use the focus knob (4) to place the sample into focus and readjust the condenser (7) and light intensity for the clearest image (with low power objectives you might need to reduce the light intensity or shut the condenser).
7. When you have a clear image of your sample with the lowest power objective, you can change to the next objective lenses. You might need to readjust the sample into focus and/or readjust the condenser and light intensity. If you cannot focus on your specimen, repeat steps 3 through 5 with the higher power objective lens in place. **Do not let the objective lens touch the slide!**
8. When finished, lower the stage, click the low power lens into position and remove the slide.





Compiled By: Atiq Rehman